

Communication Plan & Challenge Log

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Communication Plan & Challenge Log
Project Phase	Phase 8: Project Demonstration
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Team Interaction Plan

Interaction Meeting	Frequency Target	Assigned Channel Type	Core Objectives
Daily Stand-ups	Daily (Mon-Fri)	WhatsApp / Team Room	Brief sprint progress audits and dependency block checks
Sprint Planning	Once per week	Google Meet Integration	Manage backlog tasks, assign story points, review sprints
Client Review Panel	End of Sprint	Google Meet & Live Demo	Present functional features, gather feedback

2. Resolved Team Challenges Logs

Observed Challenge	Root Technical Cause	Resolution Actions Executed
Class Imbalance Bias	Approved records in training set outweighed rejected records 3:1	Applied Synthetic Minority Over-sampling Technique (SMOTE) to balance labels
Feature Scaling Mismatch	Continuous variables input variance distorted distance calculations	Created a unified StandardScaler pipeline and serialized it via joblib
Model Loading Error	Missing label translations threw runtime dictionary exceptions	Created label_encoders.pkl matching prediction key lookups